

In one line: macOS has **two keychains**: the legacy **file-based** ones (login.keychain-db, System.keychain, any file you create) searched in a **search list** and guarded by per-item **ACLs + a partition list**, and the iOS-style **data protection keychain** (one per user, guarded by **access groups** from entitlements; needed for iCloud Keychain, biometrics, Secure Enclave). codesign finds its **identity** (certificate + private key) in the *file-based* search list — so CI must create, unlock and authorise a keychain.

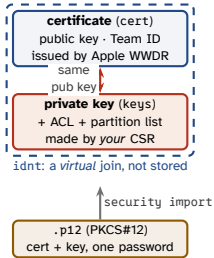
1 · search list (file-based, per user)

build.keychain-db (CI, first)
login.keychain-db – default
/Library/Keychains/System.keychain
SystemRootCertificates.keychain

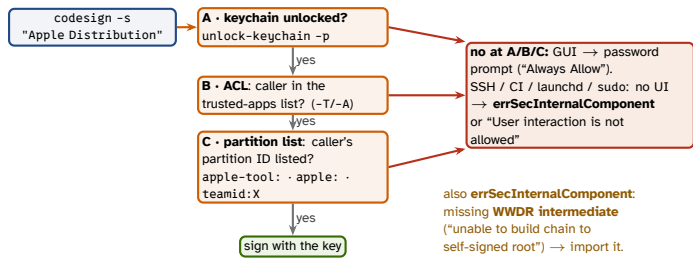
list-keychains -d user -s A B replaces the list — pass login too. login unlocks with your login password; a CI one doesn't. codesign walks the list top-down.

separate: data protection keychain
one per user · access groups · iCloud sync · *not* in the search list

2 · identity = cert + private key



3 · codesign asks for the key — three gates



also **errSecInternalComponent**:
missing **WWDR intermediate**
("unable to build chain to
self-signed root") → import it.

How it works

- File-based** = classic Mac OS heritage, "on the road to deprecation" (SecKeychainCreate deprecated, macOS 12 SDK); **data protection** = iOS's, on the Mac since iCloud Keychain (10.9) — opt in with kSecUseDataProtectionKeychain.
- Item classes** (security 4-letter codes): genp generic password, inet internet password, cert, keys; idnt = identity, assembled at query time from a cert whose public key matches a stored private key.
- ACL** = which apps may use an item without a prompt (-T /usr/bin/codesign; -A = any app, no prompt). Code identity is the app's **designated requirement**, so a re-signed or re-channelled app (e.g. TestFlight build) gets a new prompt.
- Partition list** (added later to the file-based keychain, no public API): an extra ACL tag on a key naming allowed *signer groups* — apple: (Apple-signed system code), apple-tool: (Apple tools such as codesign/security), teamid: ABCDE12345. A key imported by security import gets only apple-tool: (unverified).
- Identity**: the .cer Apple issues holds the public key only; the private key was created on the Mac that made the CSR. Export both as a .p12.

File-based vs data protection keychain

	file-based	data protection
where	files (login, System, yours) in a search list	one per user, by caller context
who may read	per-item ACL + partition list; prompts	access groups (entitlements) + SecAccessControl
sync	never	iCloud Keychain (kSecAttrSynchronizable)
daemons / root	yes (System keychain)	no — user login context only
tools	security, Keychain Access, codesign	SecItem on all Apple platforms

Remember

"Search list finds it, unlock opens it, ACL + partition list let codesign touch it."

Example — the CI keychain recipe

```

KC="$RUNNER_TEMP/build.keychain-db"; PW="$($uidgen)" # throwaway
security create-keychain -p "$PW" "$KC"
security set-keychain-settings -lut 21600 "$KC" # lock on sleep, after 6 h
security unlock-keychain -p "$PW" "$KC"
security import dist.p12 -k "$KC" -P "$P12_PW" -f pkcs12 \
-T /usr/bin/codesign -T /usr/bin/security # ACL: these tools
security set-key-partition-list -S apple-tool:,apple: -s -k "$PW" "$KC"
security list-keychains -d user -s "$KC" login.keychain-db # keep login!
security find-identity -v -p codesigning "$KC" # expect 1 valid identity
xcodesign archive ... OTHER_CODE_SIGN_FLAGS="--keychain $KC"
security delete-keychain "$KC" # always, in a trap/post step
  
```

Where the pieces come from

- Apple WWDR intermediate** must be in a searched keychain or codesign cannot build the chain — download it from Apple PKI (apple.com/certificateauthority).
- Keychain Access** edits ACLs ("Access Control" tab) of *file-based* items only; security dump-keychain -a prints the ACLs.

security — the commands you are asked about

create-keychain -p	new file keychain (.keychain-db) + password
set-keychain-settings -lut	-l lock on sleep · -u lock after -t secs; no flags = never lock
unlock-keychain -p	gate A; again in every new session (SSH)
import -k -P -T/-A -f	.p12/.cer into keychain -k; -T app into the key's ACL
set-key-partition-list -S -s -k	partitions · signing keys only · keychain password
list-keychains -d user -s	set the search list (no -s = print it)
default-keychain -s	where new items land; CI rarely needs it
find-identity -v -p codesigning	valid signing identities: SHA-1 + name
find-generic-password -s -a -w	read a secret (-w prints only it)
add-generic-password -U	store / update (-U) a secret
add-trusted-cert -r trustRoot	trust a private CA (-d: admin domain)
cms -D -i	decode CMS, e.g. a .mobileprovision
delete-keychain	remove the file + its search-list entry

Traps

- ! Fixing the ACL won't help a **locked** keychain — unlock first (Apple DTS).
- ! list-keychains -s \$KC alone **drops login** from the search list.
- ! find-identity: "0 valid identities" though the cert is there → private key missing (imported .cer, not .p12) or WWDR intermediate missing/expired.
- ! sudo codesign: root can't see your login keychain.
- ! -A on a long-lived keychain = every app reads the key silently; OK only on a throwaway CI keychain.
- ! Mac SecItem *without* kSecUseDataProtectionKeychain lands in the **file-based** keychain — ACL prompts after re-signing, no access-group sharing.
- ! security find-generic-password -w in a CI log prints the secret.